Consider the following for the next **three (03)** items that follow :

There are two points P and Q due south of a leaning tower, which leans towards north. P is at a distance x and Q is at a distance y from the foot of the tower $(x > y)$. The angles of elevation of the top of the tower from P and Q are 15° and 75° respectively.

43. At what height is the top of the tower above the ground level ?

(a) $\dfrac{x-y}{2\sqrt{3}}$

(b) $\dfrac{x-y}{4\sqrt{3}}$

(c) $\dfrac{x-y}{4}$

(d) $\dfrac{x-y}{2}$

44. If θ is the inclination of the tower to the horizontal, then what is $\cot\theta$ equal to ?

(a) $2+\dfrac{\sqrt{3}(x-y)}{x+y}$

(b) $2-\dfrac{\sqrt{3}(x-y)}{x+y}$

(c) $2+\dfrac{\sqrt{3}(x+y)}{x-y}$

(d) $2-\dfrac{\sqrt{3}(x+y)}{x-y}$

45. What is the length of the tower ?

(a) $\dfrac{x-y}{2\sqrt{3}}\sqrt{1+\left\{2+\dfrac{\sqrt{3}(x+y)}{x-y}\right\}^2}$

(b) $\dfrac{x-y}{2\sqrt{3}}\sqrt{1+\left\{2-\dfrac{\sqrt{3}(x+y)}{x-y}\right\}^2}$

(c) $\dfrac{x-y}{4\sqrt{3}}\sqrt{1+\left\{2+\dfrac{\sqrt{3}(x+y)}{x-y}\right\}^2}$

(d) $\dfrac{x-y}{4\sqrt{3}}\sqrt{1+\left\{2-\dfrac{\sqrt{3}(x+y)}{x-y}\right\}^2}$

46. What is the value of $\operatorname{cosec}\left(-\dfrac{73\pi}{3}\right)$?

(a) $\dfrac{2}{\sqrt{3}}$

(b) $-\dfrac{2}{\sqrt{3}}$

(c) 2

(d) –2

47. What is the value of

$$\cos\left(\frac{5\pi}{17}\right)+\cos\left(\frac{7\pi}{17}\right)+2\cos\left(\frac{11\pi}{17}\right)\cos\left(\frac{\pi}{17}\right) ?$$

(a) 0

(b) 1

(c) $4\cos\left(\dfrac{6\pi}{17}\right)\cos\left(\dfrac{\pi}{17}\right)$

(d) $4\cos\left(\dfrac{11\pi}{17}\right)\cos\left(\dfrac{\pi}{17}\right)$

48. $\tan\left(\frac{3\pi}{8}\right)$ का मान क्या है ?

(a) $\sqrt{2}-1$

(b) $\sqrt{2}+1$

(c) $1-\sqrt{2}$

(d) $-\left(\sqrt{2}+1\right)$

49. $\tan^{-1}\cot(\operatorname{cosec}^{-1}2)$ किसके बराबर है ?

(a) $\frac{\pi}{8}$

(b) $\frac{\pi}{6}$

(c) $\frac{\pi}{4}$

(d) $\frac{\pi}{3}$

50. त्रिभुज ABC में, $a=4,\ b=3,\ c=2$ है। $\cos 3C$ किसके बराबर है ?

(a) $\frac{7}{128}$

(b) $\frac{11}{128}$

(c) $\frac{7}{64}$

(d) $\frac{11}{64}$

51. $\cos 36° - \cos 72°$ किसके बराबर है ?

(a) $\frac{\sqrt{5}}{2}$

(b) $-\frac{\sqrt{5}}{2}$

(c) $\frac{1}{2}$

(d) $-\frac{1}{2}$

52. यदि $\sec x = \frac{25}{24}$ है और x चौथे चतुर्थांश में स्थित है तो $\tan x + \sin x$ का मान क्या है ?

(a) $-\frac{625}{168}$

(b) $-\frac{343}{600}$

(c) $\frac{625}{168}$

(d) $\frac{343}{600}$

53. $\tan^2 165° + \cot^2 165°$ का मान क्या है ?

(a) 7

(b) 14

(c) $4\sqrt{3}$

(d) $8\sqrt{3}$

54. $\sin\left(2n\pi+\frac{5\pi}{6}\right)\sin\left(2n\pi-\frac{5\pi}{6}\right)$ का मान क्या है, जहां $n \in Z$ है ?

(a) $-\frac{1}{4}$

(b) $-\frac{3}{4}$

(c) $\frac{1}{4}$

(d) $\frac{3}{4}$